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Homework 1

Question 1

Mention any 5 Hardware Vulnerabilities with examples?

Hardware vulnerabilities are a weakness that exists in the physical components and architectural design of computing systems. Unlike software vulnerabilities, these issues are usually much harder to fix once hardware is deployed. In many cases, they will likely persist for the entire operational lifetime of the device. (Tehranipoor & Bhunia, 2019). One common Hardware vulnerability involves side-channel attacks. Rather than focusing on breaking encryption directly. These attacks exploit indirect information leaked during hardware operation, such as power consumption, timing variations or electromagnetic emissions. For example, power analysis attacks can be used against smart cards to extract cryptographic keys by observing power usage during encryption operations (Tehranipoor & Bhunia, 2019).

Another significant vulnerability is hardware Trojans, which are malicious modifications to the hardware installed during the design or manufacturing stages of integrated circuits. These Trojans can create hidden backdoors or leak sensitive data such as encryption keys with the only real detection being close inspection and advanced analysis techniques (Tehranipoor & Bhunia, 2019). The Rowhammer vulnerability is an example of a memory-based hardware vulnerability that exploits the physical behavior of DRAM. By repeatedly accessing specific memory rows, attackers can induce bit flips in adjacent rows, which can potentially lead to privilege escalation

or unauthorized data modifications (Tehranipoor & Bhunia, 2019). Fault injection attacks are also a vulnerability that intentionally disrupts hardware behavior but by using techniques such as voltage glitches, clock manipulation or laser exposure. These attacks can cause cryptographic algorithms to produce faulty outputs, this allows attackers to analyze and recover secret information (Tehranipoor & Bhunia, 2019). In addition, insecure hardware interfaces, such as exposed debug or firmware access ports, pose serious security risks. For example, leaving interfaces like JTAG enabled in deployed systems can allow attackers to bypass secure boot mechanisms or extract sensitive firmware data (Tehranipoor & Bhunia, 2019).

What are the Requirements for Crypto Protocols?

At a basic level, cryptographic protocols are expected to provide CIA, confidentiality, integrity and authentication. Confidentiality ensures that only authorized parties can read protected data. While integrity ensures that the data has not been altered and authentication confirms the identities of the communicating parties (Paar & Pelzl, 2014). In addition, secure protocols should support non-repudiation, this prevents participants from denying their involvement in communications. Secure key management is also critical for cryptographic protocols which include how the keys are generated, distributed, stored and revoked. Finally, cryptographic protocols must also be resistant to common attacks that arise such as replay attacks, brute-force attacks and man-in-the-middle attacks while at the same time being efficient enough to use in real systems (Paar & Pelzl, 2014).

Question 2

Define Cryptology, Cryptography, and Cryptanalysis

Cryptography focuses on the design of algorithms and protocols to secure information, common techniques of this include encryption, digital signatures and hashing. While the art of

cryptanalysis focuses on breaking or evaluating these systems and techniques to identify weaknesses in them. Cryptology is the broader field that encompasses both cryptography and cryptanalysis, covering all aspects of secure communication and is essential for building and validating secure cryptographic systems (Paar & Pelzl, 2014).

Solve the Caesar Cipher Problem

The ciphertext 'PBZROB EXOATXOB' uses a shift key of 23. Meaning each letter will then shift 23 positions forward in the alphabet when applying the Caesar Cipher. After doing this shift the resulting plaintext is 'SECURE HARDWARE'.

Question 3

Explain Stream Cipher and Block Cipher along with their Pros and Cons.

A stream cipher is a symmetric encryption algorithm that works by generating a pseudorandom keystream. This is then combined with the plaintext data one bit or byte at a time to produce the ciphertext. Because of this design, stream ciphers are typically fast and lightweight, which makes them well suited for real-time communication systems. However, the problem that arises is that if the same keystream is ever reused or synchronization between the sender and receiver is lost, then the security of the entire communication is essentially compromised (Paar & Pelzl, 2014). On the other hand, a block cipher is a symmetric encryption algorithm that operates on fixed sized blocks of data using a single shared secret key. Each block is transformed through multiple rounds of substitution and permutation to produce the ciphertext. Block ciphers are designed in mind for strong security guarantees and form the foundation of many modern cryptographic systems. In most cases block ciphers are more robust, their only hiccup is that they require some extra setting up to be used properly and efficiently. Also, when

encrypting data that does not align exactly with the block size, padding will be required, which can add complexity and processing overhead (Paar & Pelzl, 2014).

Question 4

Explain DES and AES with examples.

DES stands for Data Encryption Standard and is a symmetric block cipher that encrypts 64-bit blocks using a 56-bit key. While DES was once widely used, particularly in banking and financial transaction systems, this changed with evolving technology and its short key length makes it vulnerable to brute-force attacks and unsuitable for modern security needs (Paar & Pelzl, 2014). The Advanced Encryption Standard (AES) was the solution to the outdated DES and is also a symmetric block cipher; however, it encrypts 128-bit blocks and supports key sizes of 128, 192 and 256 bits. In real world systems, AES is widely used in areas such as HTTPS web traffic, wireless network encryption and full disk encryption. Its strong security and efficiency are the main reasons it replaced the older standards (Paar & Pelzl, 2014).

Compare AES with DES in terms of similarity and differences.

Both DES and AES share the same fundamental design as symmetric block ciphers but they are not comparable in terms of security. DES relies on a small key size and block size which in the time of its development was sufficient, but with technological advances it is now vulnerable. AES uses larger block sizes and keys to address these issues, providing significantly stronger security and while still performing efficiently on modern hardware (Paar & Pelzl, 2014).

References

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